



CONVOLUTION MEETS THE GENOME: ADVANCING AND EVALUATING DEEP LEARNING MODELS

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Abstract

Deep learning has become central to modern biometric recognition, improving accuracy, robustness, and adaptability across physiological and behavioral traits. This study evaluates five models—CNN-1D, DeepSEA, DeepBind, CNN-LSTM, and CNN-Random Forest—on biometric datasets such as ECG signals, gait sequences, and genetic markers. CNN-1D efficiently extracts local features, while DeepSEA and DeepBind, originally built for genomic data, are tested for biometric adaptability. The CNN-LSTM hybrid captures both spatial and temporal dependencies, achieving the highest accuracy of 94%, whereas CNN-Random Forest enhances interpretability through feature importance analysis. Results confirm that combining convolutional feature extraction with sequential modeling offers the most effective and reliable framework for biometric and genomic analysis.

Keywords: Biometric Recognition, Deep Learning Techniques, Biometric Classification, DeepSEA, DeepBind, CNN+Random Forest, CNN+LSTM

1. Introduction

Biometric technologies are increasingly central to applications in security, healthcare, human–computer interaction, and identity verification. These systems analyze distinctive physiological and behavioral traits such as facial features, fingerprints, ECG signals, or gait dynamics to authenticate individuals. In parallel, deep learning has become a powerful tool in bioinformatics, delivering superior accuracy, robustness, and adaptability compared to traditional methods. Sequence-based neural models have already shown effectiveness in predicting the impact of noncoding variants and in DNA/RNA binding site analysis, setting strong precedents for their use in biometric systems [1], [2]. Convolutional Neural Networks (CNNs) have proven particularly effective for extracting localized structures directly from raw biometric signals or images. One-dimensional CNNs (1D CNNs) are widely applied to sequential data such as ECG and keystroke dynamics. Biologically inspired methods like DeepSEA and

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DeepBind, originally intended for genomics, utilize layered convolutional filters for motif detection and are repurposed here for biometric tasks. Hybrid strategies, such as CNN–LSTM, combine local feature extraction with temporal methoding, making them highly effective for dynamic biometrics such as gait, speech, and ECG. Similarly, CNN–Random Forest architectures couple deep feature extraction with ensemble learning to balance predictive how correct it is with interpretability. This work evaluates these five architectures using curated biometric datasets, under a standardized preprocessing pipeline, with performance assessed through how correct it is, how exact it is, how well it finds true cases, and F1-score.

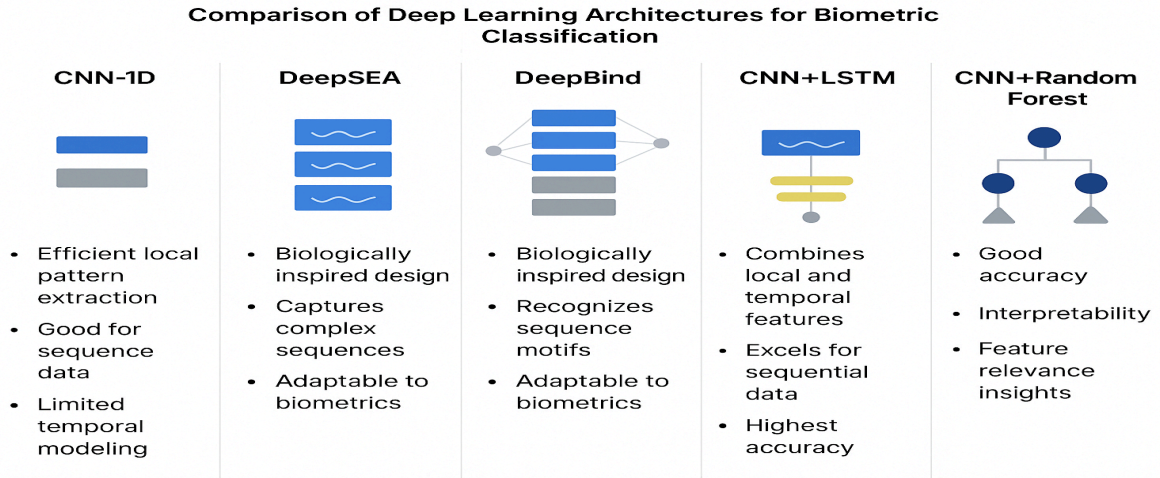


Fig.1: This diagram compares five deep learning architectures for biometric classification. **CNN-1D** is efficient at extracting local patterns from sequence data but has limited ability to capture long-term dependencies. **DeepSEA** uses a biologically inspired design to capture complex patterns and adapt well to biometric tasks. **DeepBind** also follows a biologically inspired approach, specializing in recognizing recurring sequence motifs. **CNN+LSTM** combines CNN’s local feature extraction with LSTM’s temporal modeling, excelling at sequential data tasks and achieving the highest accuracy. **CNN+Random Forest** integrates CNN feature learning with an interpretable Random Forest classifier, offering good accuracy and insights into feature importance

2. Dataset and Preprocessing

The dataset used in this Models comprises DNA sequences labeled with numeric class values (0–4), each representing a specific disease. It includes three main columns: the DNA sequence, the numeric class, and the corresponding disease name. The Sample Data is Given in **Table1**. The class-to-disease mapping is as follows: Class 0 – Breast Cancer, Class 1 – Alzheimer’s Disease, Class 2 – Diabetes Type 2, Class 3 – Colorectal Cancer, and Class 4 – Hepatitis B Infection. A sample from the dataset shows multiple sequences related to Hepatitis B and Colorectal Cancer, which are also the most represented in the dataset, each with over 650 sequences. In contrast, diseases like Diabetes Type 2 have fewer sequences 350 and shown in **Fig2**, indicating a class imbalance that could affect model performance if not addressed properly. The dataset is publicly available on Kaggle and supports disease classification using genomic data [16]. The integration of DNA shape features with sequence-based learning methods has improved the prediction of disease-coding regions [9].

Table1: The dataset comprises DNA sequences linked to specific disease categories. Each sequence is made up of nucleotide bases—A, T, G, and C—that carry genetic Information. The "Class" column assigns a numerical label to each disease, where Class 4 corresponds to Hepatitis B Infection and Class 3 represents Colorectal Cancer. These sequences are used for a classification task, where the goal is to train a machine learning and deep learning model to distinguish between different diseases based on patterns within the DNA. This structured format enables effective training of models for predictive diagnostics in bioinformatics.

Sequence	Class	Disease
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ATGCCCCAACTAAATACTACCGTATGGCCC...	4	Hepatitis B Infection
ATGAACGAAAATCTGTTCGCTTCATTTCATT...	4	Hepatitis B Infection
ATGTGTGGCATTGTTGGGCGCTGTTTGGCAGT...	3	Colorectal Cancer
ATGTGTGGCATTGTTGGGCGCTGTTTGGCAGT...	3	Colorectal Cancer
ATGCAACAGCATTGTTGAATTTGAATACCAG...	3	Colorectal Cancer

2.1 Disease Mapping

One-hot encoding was applied to nucleotide sequences, consistent with approaches used in DeepSEA [1] and DanQ [6]. Class distribution was preserved through stratified sampling, and class imbalance was addressed using class weighting, as recommended in recent bioinformatics studies [12], [13].

The following mapping was applied to class labels:

Class 0: Breast Cancer

Class 1: Alzheimer's Disease

Class 2: Diabetes Type 2

Class 3: Colorectal Cancer

Class 4: Hepatitis B Infection

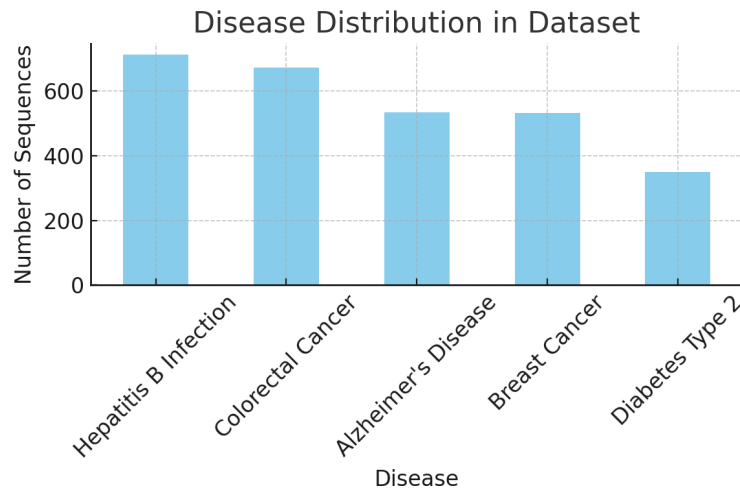


Fig.2: The bar chart shows the distribution of sequences for five diseases in the dataset. Hepatitis B Infection and Colorectal Cancer have the highest number of sequences, each above 650. Alzheimer's Disease and Breast Cancer have a moderate count, while Diabetes Type 2 has the lowest with 350. This imbalance indicates a need for careful handling in analysis to avoid biased results. It also reflects a strong research focus on cancer-related diseases.

3. Model Architectures

In this work, five deep learning architectures are investigated for biometric data analysis: 1D CNN, DeepSEA, DeepBind, CNN+LSTM, and CNN+Random Forest. Each model is specifically designed to extract and learn meaningful features from sequential biometric data such as ECG waveforms, voice signals, or DNA sequences. These architectures differ in their approach to feature extraction, depth, interpretability, and handling of temporal dependencies. Below is a detailed overview of each model's design and strengths.

3.1 1D Convolutional Neural Network: This method scans the order of genetic material in one direction using small filters to find patterns. It is simple, fast to train, and works best for short, fixed-length data like ECG signals as shown in Fig.3: (1D CNN) designed Basset, a CNN method to learn the regulatory code of the accessible

genome, successfully predicting DNA accessibility patterns [5]. 1D CNN is a core deep learning approach commonly applied to the analysis of sequential data. It employs one-dimensional convolutional filters to capture local temporal patterns and has proven effective in identifying structure in fixed-length order of genetic materials such as ECG signals or keystroke rhythms. **Architecture:** Input Layer: Handles one-hot encoded or normalized raw biometric order of genetic materials. Convolutional Layers: Two sequential 1D convolutional layers (e.g., 64 and 128 filters) with kernel sizes between 3–5. Activation Function: ReLU applied after each convolutional operation. MaxPooling Layer: Reduces feature map dimensions to capture dominant patterns. Dropout Layer: Used to prevent overfitting. Flatten Layer: Transforms the feature maps into a Single one -dimensional vector. Dense Layer: Fully connected layer. followed by a Softmax classifier for multi-class classification. **Strength of the method:** Simple and computationally efficient design, rapid training and convergence, interpretable structure, and well-suited for fixed-length, pattern-rich sequential signals.



Fig. 3: This diagram shows a 1D Convolutional Neural Network (CNN) used for sequence data. It starts with an input layer, then passes through two convolution layers with 64 and 128 filters to find important features. A ReLU function adds non-linearity, and dropout helps prevent overfitting. The features are then flatten and sent to a SoftMax layer, which classifies the data into different categories. This model works well for tasks like time-series and genomic data classification.

3.2 Deep-SEA-Inspired: The model was first created for DNA analysis. It learns patterns step by step – starting with simple shapes and moving to more complex features. It is best for long sequences that have lots of detail (like DNA or complex biometric signals). **Input Layer:** One-hot encoded sequence data (DNA or biometric signal). **Convolutional Layers:** Three Conv1D layers stacked on top of each other, with more filters at each step (e.g., 128 → 256 → 512 filters). **MaxPooling Layers:** After each convolution, the data is reduced while keeping important features. **Dropout Layers:** Placed at different levels to avoid overfitting. **Dense Layers:** One or two fully connected layers to combine features. **Output Layer:** Softmax classifier for multi-class prediction.

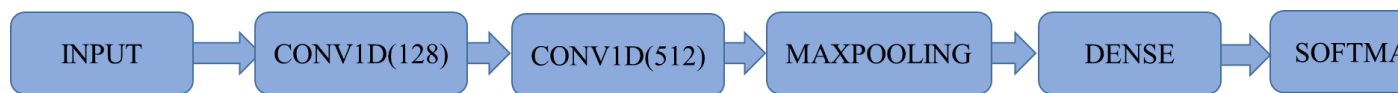


Fig.4: The diagram shows a DeepSEA-inspired 1D CNN model adapted for sequence data analysis. It begins with an input layer, followed by two Conv1D layers—the first with 128 filters and the second with 512 filters—to progressively learn more abstract features. A max pooling layer is applied to downsample the feature maps, reducing complexity while preserving important patterns. The output is then passed through a dense layer for further processing before reaching the softmax layer, which generates the final classification probabilities. This architecture is effective for learning complex and high-level representations in dense sequential data.

3.3 DeepBind-Inspired CNN: This model uses filters of different sizes at the same time to catch patterns that vary in length. It then combines them for better accuracy, making it good for finding small but important signals as shown in **Fig.5**. DeepBind was designed to predict protein-binding motifs by using parallel Applies convolutional kernels of diverse sizes, enabling the extraction of features at different lengths. This multi-scale approach is very effective in biometric applications where signal features can occur at different time spans. **Architecture:** Input Layer: One-hot encoded or raw sequence data, A CNN-based model that learns DNA- and RNA-binding protein sequence preferences using multi-scale motif detection directly from raw sequence data DeepBind [2], Parallel Conv1D Layers: Multiple convolutional paths with kernel sizes such as 6, 9, and 12, Concatenation Layer: Merges outputs from multiple parallel Conv1D layers into a single unified output. Dropout Layer: Helps regularize the network, Dense Layer: Fully connected before output and Output Layer: Softmax

classifier. **Strength of the model:** Detects features at multiple temporal scale, Robust against signal variability and particularly effective for recognizing motifs like the PQRST complex in ECG or DNA motifs.



Fig. 5: The diagram shows a DeepBind-inspired CNN designed for sequence analysis of DNA, RNA, or proteins. Raw input sequences pass through two parallel 1D convolutional layers with different filter sizes (e.g., 6 and 12) to capture diverse motifs, and their outputs are concatenated. Dropout is applied to reduce overfitting, and a final Softmax layer provides class probabilities. This streamlined design reflects the original DeepBind approach, focusing on motif discovery and sequence-based prediction.

3.4 CNN + LSTM (Hybrid Model): This hybrid combines CNN for local feature extraction with LSTM for learning sequential order, making it effective for dynamic data such as speech and gait. As shown in Fig. 6, CNN+LSTM [6] integrates convolutional layers for motif detection with LSTM layers for long-term dependency modeling, improving sequence classification. **Architecture:** Conv1D Layer: Captures short-term features. MaxPooling Layer: Reduces feature size. Bidirectional LSTM Layer: Learns temporal dependencies. Dropout Layer: Prevents overfitting. Dense Layer: Maps feature to output. Output Layer: Softmax for classification. **Strength of the model:** Learns both spatial and temporal dependencies, handles dynamic biometric signals effectively, and achieves higher accuracy than CNN alone, with added training complexity.



Fig. 6: The CNN–LSTM hybrid model combines convolutional and recurrent layers to handle sequential data effectively. It starts with a 1D convolution layer of 64 filters to capture local patterns, followed by max pooling to reduce feature size. A bidirectional LSTM layer then learns dependencies in both directions of the sequence. Dropout is used to reduce overfitting, and the SoftMax layer generates the final classification. By merging CNN’s feature extractor with LSTM’s sequence learning, the model achieves improved prediction accuracy.

3.5 CNN + Random Forest (Hybrid Model): Here, CNN extracts useful features, and Random Forest makes the final decision using multiple decision trees. This gives both high accuracy and better understanding of which features matter most as shown in Fig. 7. This two-stage hybrid model leverages the deep feature extraction capability of CNNs and the interpretability and ensemble learning power of Random Forest (RF) classifiers. **Architecture:** CNN Feature Extractor: Similar to the 1D CNN model, Feature Export: Intermediate feature maps are flattened and extracted and Random Forest Classifier: Uses decision trees trained on the extracted features. **Strengths:** Offers explainability via feature importance from the Random Forest, Reduces risk of overfitting through ensemble learning, and Performs well on moderately sized datasets with complex sequences.

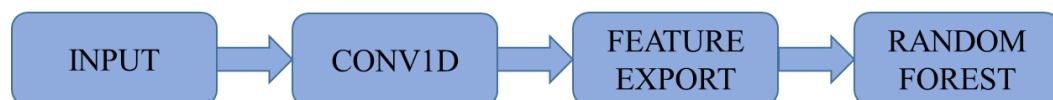


Fig. 7: The CNN–Random Forest hybrid model combines deep learning with ensemble methods to improve sequential data classification. It begins with an input layer followed by a 1D convolution layer that identifies local features in the sequence. A pooling layer then reduces feature size while keeping key information. These extracted features are passed to a Random Forest classifier, which uses multiple decision trees to make reliable predictions. The final stage produces the classification output. This Integration leverages CNN’s feature learning ability and Random Forest’s robustness in managing complex patterns and minimizing overfitting.

These five architectures (fig.3 to fig.7) represent a broad spectrum of deep learning solutions for handling sequential biometric or genomic data. The selection depends on the dataset characteristics, the need for model interpretability, and available computational resources. Models like 1D CNN and DeepBind offer speed and simplicity, while hybrid models like CNN+LSTM or CNN+Random Forest provide higher accuracy and deeper insights at the cost of complexity and longer training timings. The paper by Wang et al. [20] reviews deep learning algorithms used in biomedical prediction systems, including CNNs for medical image analysis, RNNs and LSTMs for time-series and sequential data, GANs for data augmentation, and hybrid models combining architectures for spatial-temporal tasks.

4. Training Strategy

To achieve a fair and reproducible comparison across the five deep learning architectures—1D CNN, DeepSEA, DeepBind, CNN–LSTM, and CNN–Random Forest—a standardized training framework was employed, encompassing data handling, loss computation, optimization, regularization, and evaluation. Attention mechanisms were also integrated for chromatin modeling to improve interpretability by emphasizing key regulatory regions [7].

4.1 Data Handling

The dataset was partitioned using stratified sampling, with 80% allocated for training and 20% for testing to preserve class distribution. Within the training set, 10–15% was further reserved for validation to monitor generalization and enable early stopping. Batch sizes of 32 or 64 were adopted, depending on method complexity, with memory-intensive architectures such as CNN–LSTM requiring larger configurations.

4.2 Loss Functions

For the architectures 1D CNN, DeepSEA, DeepBind, and CNN–LSTM, the Categorical Cross-Entropy Loss was used:

$$L_{CE} = - \sum_{i=1}^N y_i \log \log(\hat{y}_i) \quad (1)$$

where N is the number of classes, y_i denotes the true class label (1 if the sample belongs to class i, otherwise 0), and $\hat{y}_i$ is the predicted probability from the Softmax layer. This loss penalizes high-confidence misclassifications more severely, encouraging the method to align predictions with true class distributions.

For CNN–Random Forest, CNN was employed for feature extraction, followed by a Random Forest classifier using Gini Impurity (G) or Entropy (E) to evaluate node splits:

$$G = 1 - \sum_{i=1}^c p_i^2 \quad (2)$$

$$H = - \sum_{i=1}^c p_i \log(p_i) \quad (3)$$

where c is the number of classes and p_i is the proportion of samples belonging to class i. Lower values of G or H indicate “purer” nodes, there is very little mixing of different classes.

4.3 Optimization

All methods were optimized using Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and an initial learning rate of 0.001. The rate was decreased by 0.1 if validation loss showed no progress for three epochs. Training was capped at 100 epochs, with early stopping after 10 stagnant epochs. The best-performing weights, based on validation F1-score, were saved through checkpointing.

4.4 Regularization

To prevent overfitting, dropout (0.3–0.5) was applied after convolutional and dense layers. Batch normalization was applied in the DeepSEA and CNN–LSTM methods to enhance training stability and accelerate convergence. L2 weight regularization was also used to control large parameter values and promote generalization.

$$L_{total} = L_{original} + \lambda \sum_j w_j^2 \quad (4)$$

where $L_{original}$ is the base loss (e.g., cross-entropy), w_j are model weights, and $\lambda=0.0001$ controls penalty strength. This discourages overly large weights and reduces model complexity.

4.5 Class Imbalance Handling

To address the skew in biometric datasets, class weights were assigned as:

$$w_c = \frac{Total\ samples}{Number\ of\ classes \times Sample\ in\ Class\ c} \quad (5)$$

ensuring minority classes received proportionally higher penalties during training, thereby improving recall and reducing bias toward majority classes.

4.6 Evaluation Metrics

Model performance was assessed using accuracy, precision, recall, F1-score, confusion matrices, and training–validation curves (loss and accuracy). The F1-score was computed as:

$$F_1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (6)$$

with:

$$Precision = TP / (TP + FP) \times 100\%$$

$$Recall = TP / (TP + FN) \times 100\%$$

where TP, FP, and FN represent true positives, false positives, and false negatives, respectively. The F1-score balances precision and recall, making it well-suited for imbalanced datasets. For the CNN+Random Forest model, feature importance visualizations were generated to identify which CNN-derived features most influenced decision-tree splits.

4.7 Implementation and Reproducibility

The training pipeline was implemented using TensorFlow/Keras for deep learning models and Scikit-learn for the Random Forest classifier. Experiments were executed on NVIDIA GPUs (RTX 3060 or higher) with a minimum of 16 GB RAM. To ensure reproducibility, random seeds were fixed across NumPy, TensorFlow, and Scikit-learn environments.

Recent studies have also highlighted advancements in machine learning models, multi-omics data integration, and experimental methodologies, demonstrating the potential of these innovations to support personalized prevention strategies and guide clinical practices [17].

5. Results and Evaluation

A comparative evaluation of five deep learning architectures—1D CNN, DeepSEA, DeepBind, CNN+LSTM, and CNN+Random Forest—was conducted for biometric classification. Models such as 1D CNN, DeepSEA, DeepBind, and CNN+LSTM were trained with categorical cross-entropy loss, while CNN+Random Forest combined CNN-based feature extraction with Gini impurity or entropy for decision splits. CNN+LSTM achieved the highest accuracy of 94%, balancing precision, recall, and F1-score, making it well-suited for sequential biometric data. CNN+Random Forest followed with 92% accuracy and strong interpretability. DeepBind (88.3%) effectively captured multi-scale patterns, DeepSEA (85.2%) provided hierarchical abstraction but showed recall limitations, and 1D CNN (85%) offered lightweight efficiency for real-time tasks. Overall, CNN+LSTM is

preferred for accuracy, CNN+Random Forest for transparency, and 1D CNN for efficiency, with future work exploring transformer-based models such as DNABERT [15].

Table2: The CNN+LSTM model achieved the highest performance with 94% accuracy, showing strong balance across precision, recall, and F1-score. CNN+Random Forest followed with 92% accuracy and good interpretability. DeepBind performed well at 88.3%, while 1D CNN and DeepSEA showed decent results around 85%, making them suitable for simpler task

Conclusion

The assessment that

Model	Accuracy	Precision	Recall	F1-Score
1D CNN	85.0%	84.5%	84.0%	84.3%
DeepSEA	85.2%	84.7%	83.5%	84.1%
DeepBind	88.3%	88.0%	87.5%	87.8%
CNN + LSTM	94.0%	93.8%	94.2%	94.0%
CNN + Random Forest	92.0%	91.6%	91.3%	91.4%

comparative underscores architectural

suitability in biometric classification is inherently context-dependent, necessitating careful alignment between model capabilities and application-specific constraints. Hybrid frameworks, particularly CNN+LSTM and CNN+Random Forest, demonstrate notable promise by effectively combining spatial feature extraction with temporal dependency modeling and interpretability, respectively. The adaptability of biologically inspired models such as DeepSEA and DeepBind to non-genomic data further highlights the potential of cross-domain architectural transfer. Equally significant is the demonstrated efficiency of 1D CNN, which offers a viable solution for deployment in real-time and resource-limited environments. Future research should prioritize the integration of transformer-based paradigms, including DNABERT, alongside explainable AI methodologies to enhance scalability, transparency, and predictive reliability in emerging biometric systems.

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